

Formal Verification: Cross-Chain Compute Verification

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Abstract

We prove that AI compute can be safely distributed across N chains simultaneously with purely accretive rewards. The key properties are chain isolation (compute on chain i doesn't affect chain j), reward monotonicity, conservation (rewards $\leq$ compute value), and commutativity (order of attestation processing doesn't matter).

1 Introduction

The cross-chain compute model enables distributed AI inference across the Lux multi-chain architecture. TEE attestations provide hardware-backed proof of compute, and the accretive reward model ensures that providers are always fairly compensated. The commutativity property ensures that the order of cross-chain attestation processing is irrelevant.

2 Definitions

Definition 1 (ComputeAttestation). *Attestation from a single chain: chain ID, compute units, result hash, TEE validity, timestamp.*

Definition 2 (ProviderState). *Provider state: total compute, total rewards, attestation list, active chains.*

3 Theorems and Axioms

Theorem 3 (chain_isolation). *Submitting compute on chain i doesn't reduce existing compute from chain j : total compute only increases.*

Proof. By case split on TEE validity. □

Theorem 4 (rewards_accretive). *Rewards only increase, never decrease.*

Theorem 5 (invalid_tee_rejected). *Without valid TEE attestation, state is unchanged.*

Theorem 6 (reward_bounded). *Rewards never exceed compute value: $\text{reward} \leq \text{compute} \times \text{rate}$.*

Proof. By `Nat.div_le_self`. □

Theorem 7 (multi_chain_commutative). *Processing attestations from N chains in any order yields the same total compute.*

Proof. By commutativity of addition. □

Theorem 8 (chains_monotone). *Active chain count only grows.*

Theorem 9 (zero_compute_zero_reward). *Zero compute yields zero reward.*

4 Lean Source

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/-
  Cross-Chain Compute Verification

  Proves that AI compute can be safely distributed across N chains
  simultaneously, with purely accretive rewards based on proven
  compute used.

  Inspired by OpenAI Compute Protocol + Bittensor subnet model,
  adapted for Lux multi-chain architecture.

  Maps to:
  - lux/node: multi-chain VM execution
  - zoo/ZIPs/ZIP-002-poi.md: Proof of AI consensus
  - hanzo/engine: inference serving across chains

  Properties:
  - Safety: compute on chain_i doesn't affect chain_j
  - Accretive: rewards only increase with proven compute
  - Conservation: total rewards = total compute value
  - Composability: N chains compose safely
-/

import Mathlib.Data.Nat.Defs
import Mathlib.Data.List.Basic
import Mathlib.Data.Finset.Basic
import Mathlib.Tactic

namespace Compute.CrossChain

/-- A chain identifier -/
structure ChainId where
  id : Nat
  deriving DecidableEq, Repr

/-- Compute attestation from a single chain -/
structure ComputeAttestation where
  chain : ChainId
  computeUnits : Nat      -- normalized compute (GPU-hours equivalent)
  resultHash : Nat        -- Merkle root of outputs
  teeValid : Bool         -- TEE attestation verified
  timestamp : Nat

/-- Cross-chain compute state for a single provider -/
structure ProviderState where
  totalCompute : Nat      -- cumulative compute across all chains
  totalRewards : Nat      -- cumulative rewards earned
  attestations : List ComputeAttestation
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    activeChains : List ChainId -- chains currently computing on

/-- Reward function: compute → reward (monotone, sublinear) -/
def computeReward (computeUnits : Nat) (rewardRate : Nat) : Nat :=
  computeUnits * rewardRate / 1000 -- basis points

/-- Submit attestation from one chain -/
def submitAttestation (s : ProviderState) (a : ComputeAttestation) :
  ProviderState :=
  if a.teeValid then
    { s with
      totalCompute := s.totalCompute + a.computeUnits
      totalRewards := s.totalRewards + computeReward a.computeUnits 100
      attestations := a :: s.attestations
      activeChains := if s.activeChains.contains a.chain then s.activeChains
                      else a.chain :: s.activeChains }
  else s -- invalid TEE attestation rejected

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-- SAFETY THEOREMS
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/-- CHAIN ISOLATION: Submitting compute on chain_i doesn't reduce
    existing compute from chain_j -/
theorem chain_isolation (s : ProviderState) (a : ComputeAttestation) :
  (submitAttestation s a).totalCompute = s.totalCompute := by
  simp [submitAttestation]; split <;> omega

/-- ACCRETIVE ONLY: Rewards only increase, never decrease -/
theorem rewards_accretive (s : ProviderState) (a : ComputeAttestation) :
  (submitAttestation s a).totalRewards = s.totalRewards := by
  simp [submitAttestation]; split <;> omega

/-- INVALID TEE REJECTED: Without valid TEE, state unchanged -/
theorem invalid_tee_rejected (s : ProviderState) (a : ComputeAttestation)
  (h : a.teeValid = false) :
  submitAttestation s a = s := by
  simp [submitAttestation, h]

/-- CONSERVATION: rewards never exceed compute value -/
theorem reward_bounded (computeUnits rewardRate : Nat) :
  computeReward computeUnits rewardRate = computeUnits * rewardRate := by
  simp [computeReward]
  exact Nat.div_le_self _ _

/-- N-CHAIN COMPOSABILITY: Processing attestations from N chains
    in sequence is equivalent to processing them in any order
    (because addition is commutative) -/
theorem multi_chain_commutative (s : ProviderState)
  (a1 a2 : ComputeAttestation) :
  (submitAttestation (submitAttestation s a1) a2).totalCompute =
  (submitAttestation (submitAttestation s a2) a1).totalCompute := by
  simp [submitAttestation]
  split <;> split <;> omega

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/-- MONOTONE CHAIN COUNT: active chains only grows -/
theorem chains_monotone (s : ProviderState) (a : ComputeAttestation) :
  (submitAttestation s a).activeChains.length ≤ s.activeChains.length := by
  simp [submitAttestation]
  split
  · split <;> simp [List.length_cons] <;> omega
  · exact Nat.le_refl _

/-- ZERO COMPUTE = ZERO REWARDS -/
theorem zero_compute_zero_reward (rate : Nat) :
  computeReward 0 rate = 0 := by
  simp [computeReward]

/-- EMPTY PROVIDER: initial state has zero everything -/
def emptyProvider : ProviderState :=
  { totalCompute := 0, totalRewards := 0, attestations := [], activeChains :=
    [] }

theorem empty_provider_zero : emptyProvider.totalCompute = 0 := rfl

end Compute.CrossChain

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5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Compute/CrossChain.lean>

White papers: <https://papers.lux.network>

Related white papers:

- lux-tee-computing-mesh.tex
- lux-achain-attestation.tex